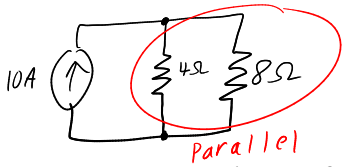
EE 2010
Fall 2026

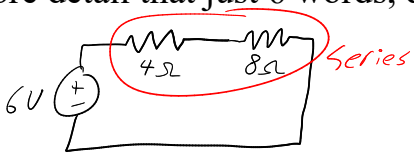
Name Wyatt Evans

Game 48
11/4/2026

Today we will talk about equivalent impedance circuits. Impedance in AC circuits is analogous to resistance in DC circuits. But first,

a) What is the definition of series components — components wired to feed into the next
b) What is the definition of parallel components — components starting and ending at the same points
*Provide more detail that just 6 words, draw examples




Impedance for each component is shown below:

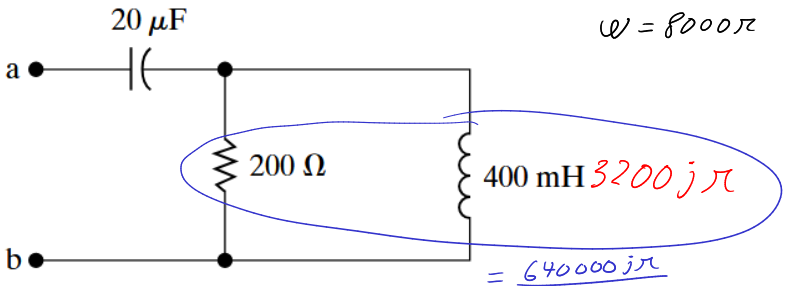
$$Z_R = R \qquad Z_L = j\omega L \qquad Z_C = \frac{-j}{\omega C}$$

Equivalent impedance (parallel and series) simplifies the exact same way as equivalent resistors…

Find the equivalent impedance of these circuits at 4000 Hz:

$4000 = \frac{\omega}{2\pi}$

$\omega = 8000\pi$

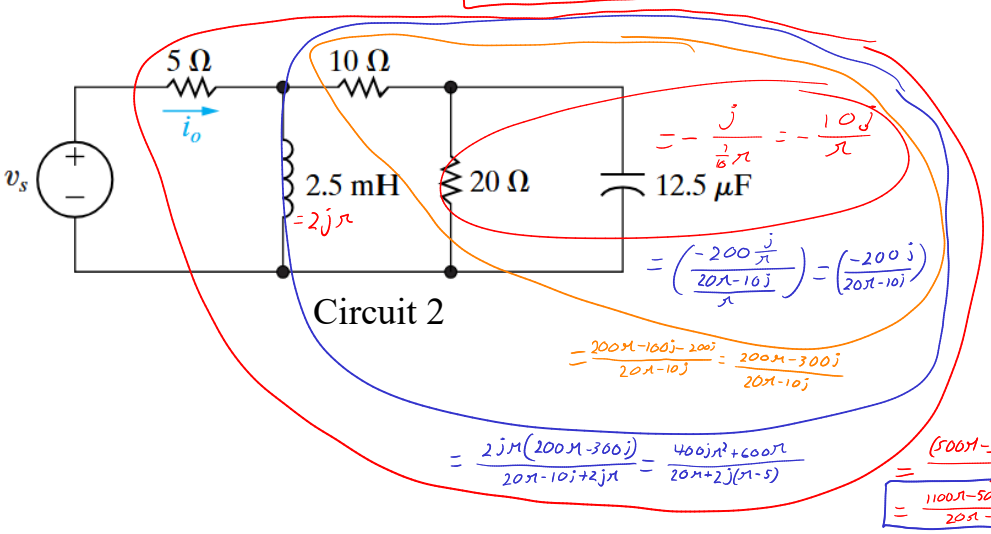



$-\frac{j}{0.16\pi} = \frac{-25\pi j}{4}$

$3200j\pi$

$= \frac{640000j\pi}{3200j\pi + 200}$

$Z_{eq} = \frac{640000j\pi}{3200j\pi+200} - \frac{25\pi j}{4}$




$= -\frac{j}{\frac{1}{16}\pi} = -\frac{16j}{\pi}$

$= \left(\frac{-200\frac{j}{\pi}}{20\pi - 16j}\right) = \left(\frac{-200j}{20\pi - 16j}\right)$

$= \frac{200\pi - 100j - 200j}{20\pi - 16j} = \frac{200\pi - 300j}{20\pi - 16j}$

$= \frac{2j\pi(200\pi - 300j)}{20\pi - 16j + 2j\pi} = \frac{400j\pi^2 + 600\pi}{20\pi + 2j(\pi - 8)}$

$= \frac{(5000\pi - 50j + 16j\pi) + (400j\pi^2 + 600\pi)}{20\pi + 2j(\pi - 8)}$

$= \frac{1100\pi - 50j + 16j\pi + 400j\pi^2}{20\pi - 16j + 2j\pi}$